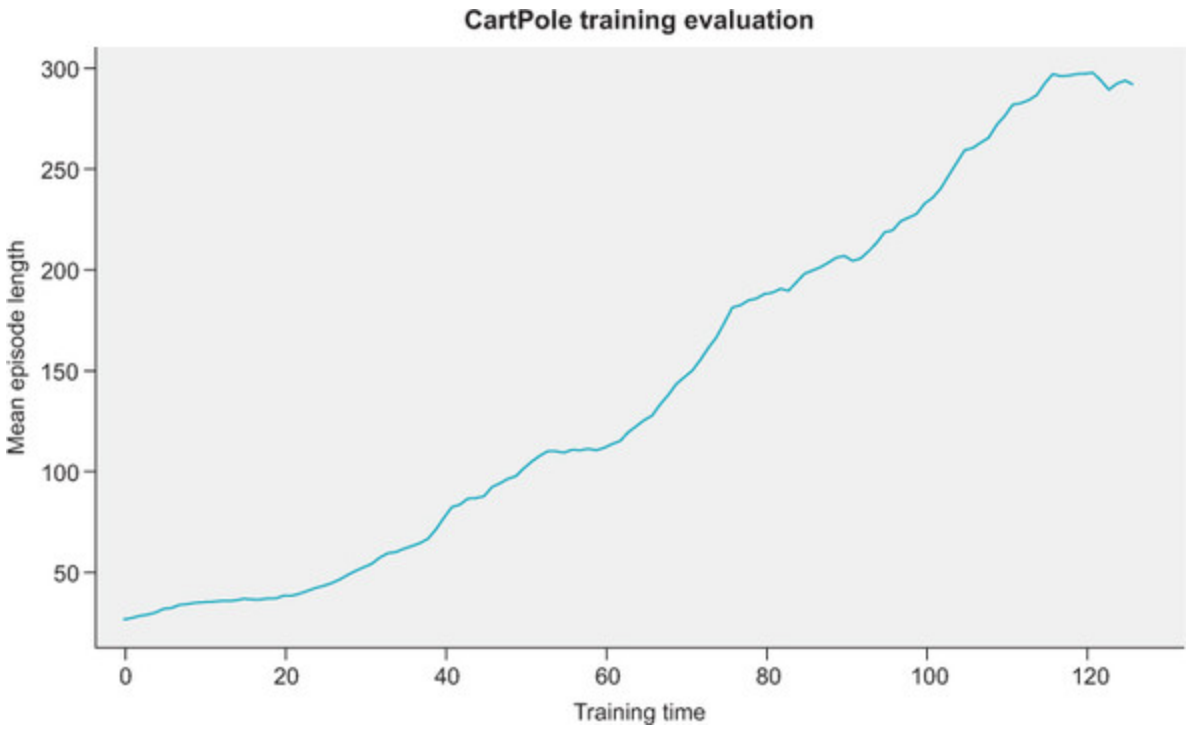
Adversarial training like this is a very powerful technique in many areas of machine learning, not just reinforcement learning. For example, generative adversarial networks (GANs) are an unsupervised method for generating realistic-appearing synthetic samples of data from a training data set using a pair of models that function similarly to an actor and critic. In fact, we will build an even more sophisticated adversarial model in chapter 8.

The take-home here is that if you're using an adversarial model, the loss will be largely uninformative (unless it goes to 0 or explodes toward infinity, in which case something is probably wrong). You have to rely on actually evaluating the objective you care about, which in our case is how well the agent is performing in the game. Figure 5.14 shows the plot of average episode length during the first 120 epochs (about 45 seconds of) training.

Figure 5.14. The mean episode length over training time for our Monte Carlo distributed advantage actor-critic model. This model is not considered a true critic, since the critic is not bootstrapping during training. As a result, the training performance has high variance.



5.4. N-step actor-critic